

S.No. : 431

NCS 4201

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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43208

Roll
No.

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B. Tech. Examination, 2024-25

(Even Semester)

PROGRAMMING CONCEPTS WITH PYTHON

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

(a) , What is a dictionary in Python? Give an example.

(b) , Ekxplain the difference between 'append' and 'extend' in Python.

(c) , What will be the output of the following code :

```
list = ['m', 'o', 'n', 'k', 'e', 'y']
```

```
print = ("@.join (list))
```

- (d) Name any two file access modes in Python.
- (e) What is object-oriented programming in Python? Give an example.
- (f) What do you mean by pow ()?
- (g) What will be the output of the following Python code :

```
def count1(s):  
    Vowels="AEIOUaeiou"  
    Count = 0  
    for C in S :  
        if c in vowels :  
            count += 1  
    return count  
print(count1('I love India'))
```

- (h) What is IDE in Python? Name any two IDEs.

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Demonstrate five different built in functions used in the string. Write a program to check whether a string is a palindrome or not?

- (b) , What are logical and Boolean operators? List all logical and Boolean operators in Python and explain the use of these operations with suitable example.
- (c) Illustrate through examples, how we can implement different type of inheritance in Python? Explain the static, instance and class method through example.
- (d) Explain expression evaluation and float representation with example. Write a Python program for how to check if a given number is fibonacci number?

SECTION – C

Note :- Attempt all questions. Attempt any two parts from each questions.

$$8 \times 5 = 40$$

- 3. (a), Discuss file I/O in Python. Explain exception handling with syntax.
- (b) , What do you mean by operator precedence and associativity? Explain.

- (c) What is the significance of module in Python?
What are the methods of importing a module?
Explain with suitable example.

4. (a) Write short notes on the following :

- (i) The programming cycle for Python
- (ii) Type conversion

(b) Create a function `min_max ( )` that takes a number as list argument and return the smallest and largest numbers.

•(c) Explain for and while loops with a flow diagram, syntax and suitable example.

5. (a) Write Python program to swap two numbers without using intermediate variables.

(b) Explain the list comprehension with any suitable example. Write a program to accept the string which contain all vowels.

(c) Explain the continue, break and pass statement with a suitable example.

6. (a) Illustrate unpacking sequences and mutable sequences with example.
- (b) Write a program to calculate the reverse of any entered number.
- (c) Explain the concept of Lambda function in Python. Write a lambda function that takes a list of numbers and returns the sum of all the numbers in the list.
